A New Fatigue Life Model for Rolling Bearings

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This paper describes a novel model for the prediction of fatigue life in rolling bearings. Central to this model is the postulation of a statistical relationship between the probability of survival, the fatigue life, and a stress-related fatigue criterion level above a fatigue limit for an elementary volume of material in the bearing. Using this concept, the stress volume to fatigue and the fatigue life of the bearing can be calculated for different loads, material and operating conditions. Comparisons between experimentally obtained rolling bearing fatigue lives and lives predicted using this theory indicate its ability to account for phenomena hitherto excluded from fatigue life predictions. Furthermore, comparisons between experimentally obtained fatigue lives for other specimens used in structural fatigue tests and fatigue lives predicted using the new model show good agreement.

Introduction

According to the commercial literature published by ball and roller bearing manufacturers, in general supported by international and national standards; e.g., ISO [1], ANSI, DIN, etc., even under ideal conditions of operation the endurance of rolling bearings in any application is ultimately terminated by fatigue. The standards specify methods of calculating basic dynamic load rating and rating life of rolling bearings well-manufactured from good quality steel, and basically of conventional design as regards the shape of rolling contact surfaces. The methods contained in the standards are based on the work performed by G. Lundberg and A. Palmgren [2, 3], conducted during the late 1930's and early 1940's. The ball and roller bearing load ratings and life calculation methods were, and remain, representative of the manufacturing methods, materials, lubricating methods and testing methods then available. Moreover, the Lundberg-Palmgren effort suffered from lack of precise knowledge of the mechanics of lubrication of concentrated contacts. In 1949, Grubin et al. [4] disclosed the probable existence of a lubricant film which could separate surfaces in rolling "contact," which otherwise were in apparent metal-to-metal contact. Grubin's revelation, confirmed experimentally by Sibley et al. [5] in 1960, did not undergo serious consideration by the manufacturers and/or users of rolling bearings until the late 1960's. Subsequently, the standards and manufacturers' catalogues accommodated this phenomenon by inclusion of variable life adjustment factors to be applied to fatigue lives calculated according to the original standards. Life adjustment factors were also included simultaneously to accommodate improvements in bearing steel obtained since the publication of the original standards. This situation has continued through the 1970's and into the 1980's.

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Notwithstanding that the Lundberg-Palmgren work was a substantial step forward in the development of rolling bearing technology, owing to the limitations in available tribological technology, certain key phenomena remained either insufficiently defined or not explained. Perhaps the most significant of these has been the inability to relate fatigue of elemental surfaces in rolling contact to structural fatigue, or even to rolling contact fatigue in ball and roller bearings. S-N (stress versus number of cycles to fatigue) curves, typical of structural fatigue for many materials in reversed bending or torsion, see Fig. 1, and having fatigue limits, have not been demonstrated to be applicable to rolling bearing fatigue. Therefore, rolling bearing applications have been burdened with a theoretical finite life under all conditions of operation. Many modern rolling bearing applications have refuted this stated limitation; e.g., recent data [6] published on endurance tests of heavily-loaded 6309 deep groove ball bearings of standard design but accurately manufactured from "clean," homogeneous steel, have demonstrated that virtually infinite fatigue life can be a practical consideration in some rolling bearing applications.

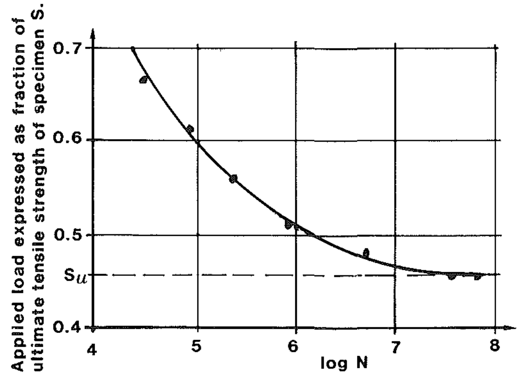


Fig. 1 An example of an S-N curve

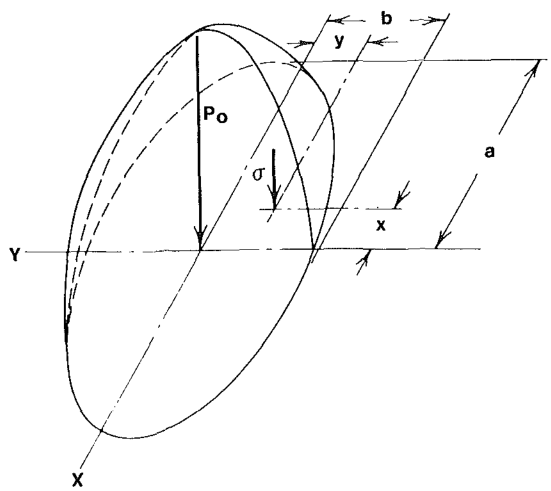


Fig. 2 Ellipsoidal surface compressive stress distribution of point contact

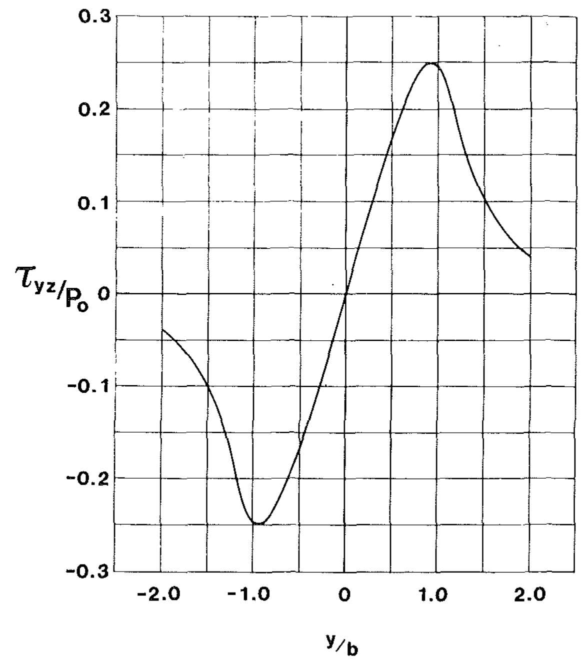


Fig. 3 τ_{yz}/p_0 **versus** y/b **for** $b/a = 0$ **and** $z = z_0$

From a material structure point of view, there does not seem to be any logical justification to differentiate between structural fatigue initiation and rolling contact fatigue initiation. It is apparent that material commences to fail when the stress at a given location overcomes the material's ability to withstand the stress. That the internal stress fields and energy dissipation in structural bending or torsion are different from those in rolling contact is certain. What is required is to define the stress or energy levels the material cannot withstand on a micro-scale basis and relate these to the stress fields and energy levels developed in applications. This will be attacked in an empirical manner in this paper.

The Lundberg-Palmgren fatigue theory presumes that a fatigue crack is initiated at a distance below the surface in rolling contact, consistent with the simultaneous occurrence at that distance of a large magnitude orthogonal shear stress and a weak point in the material. Such weak points are presumed to be stochastically distributed throughout the material. The entire subsurface stress distribution is assumed to be a function only of the normal stress which occurs between the surfaces in rolling contact, as shown by Fig. 2. Figure 3 illustrates the variation of the orthogonal shear stress, both with depth and with distance, in the direction of rolling motion. The effect on subsurface stresses of surface shear stresses, in addition to the normal stress, was not considered by Lundberg-Palmgren. Several researchers [7, 9] have subsequently defined the subsurface stress distribution which results from the combined application over a concentrated contact area of a simple normal stress and simple shear stress acting on the surface. None, however, have rigorously related this effect to fatigue life. Tallian [10] has chosen to identify competing modes of fatigue failure; i.e., surface-initiated and subsurface initiated. A rigorous solution requires the possibility of failure at any point in the material, from the surface downwards, in accordance with the applied stresses and other boundary conditions and the material surface and subsurface properties. Further, the surface stress condition is generally not simple since the contact surfaces are not portions of perfect geometrical entities such as spheres, ellipsoids, toroids, etc. These deviations from geometrical and material perfection were much more valid for the ball and roller bearings tested by Lundberg-Palmgren to confirm their theory than for modern rolling bearings.

The basic equation in the Lundberg-Palmgren theory states that for a bearing ring subjected to a number of cycles N of repeated concentrated stress, the probability of survival from subsurface-initiated fatigue is given by:

Nomenclature

a = contact ellipse major semiaxis mm
A = constant
b = contact ellipse minor semiaxis mm
c = stress criterion exponent
C = bearing basic dynamic load rating N
e = life exponent
$f(I)$ = probability density function
$g(\sigma_u)$ = probability density function
h = depth exponent
$H(x)$ = step function
I = stress-related integral in life calculation
N = fatigue life
P = bearing equivalent radial load N
p_0 = maximum Hertzian pressure N/mm^2
$P\{x\}$ = probability of occurrence of event x
R_c = cut-out radius of rotating beams mm
R_0 = minimum radius of rotating beams mm
$R_{\sigma u}(d)$ = auto-correlation function of σ_u with distance
$S(\Delta S)$ = probability of survival
$V(\Delta V)$ = volume mm^3
z_0 = depth of maximum Hertzian shear stress τ_{yz} mm
z' = stress weighted average depth mm
$\delta_D(x)$ = Dirac function
Λ = film thickness/roughness ratio
μ = coefficient of friction
σ = value of stress-related fatigue criterion N/mm^2
σ_u = fatigue limit value of N/mm^2
σ_0 = maximum normal stress occurring in a rotating beam N/mm^2
σ_{VM} = von Mises equivalent stress N/mm^2

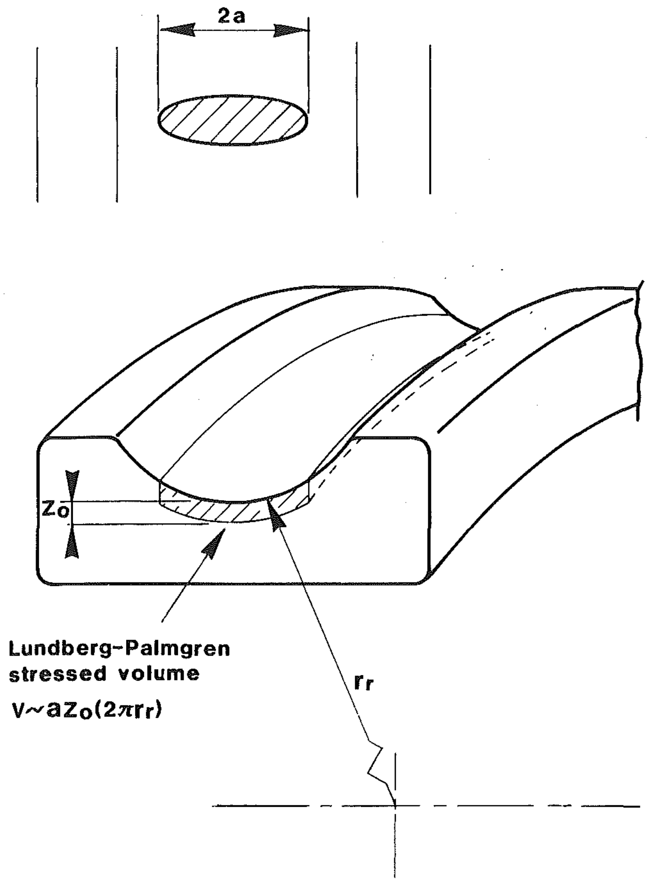


Fig. 4 Volume in fatigue risk V in reference [2] in proportion to az_0l, as shown ($l = 2\pi r_r$)

$$\ln\frac{1}{S} \sim \frac{N^e \tau_0{}^c V}{z_0{}^h} \tag{1}$$

in which τ_0 is the maximum orthogonal shear stress and z_0 is the depth at which it occurs. The stressed volume in equation (1) is assumed by Lundberg-Palmgren to be proportional to the product of the contact area semilength, the circumference of the rolling body and z_0; see Fig. 4. This proportionality is satisfactory only if the surfaces are geometrically perfect and a simple normal stress occurs between contacting surfaces. In the event of a significant surface shear stress, for example, the strong tendency for surface-initiated fatigue failure is completely ignored.

Finally, the Lundberg-Palmgren theory does not explicitly account for the rate at which energy is applied to the surfaces in rolling contact, nor for the bearing operating temperature. Lives are considered in millions of revolutions; bearing operational speeds only serve to convert predicted endurances to time values. Moreover, the development of microstructural alterations and of residual stresses below the raceways, induced by the rolling contact, have now been extensively reported; e.g., reference [11]. Perhaps such effects can be accommodated satisfactorily by considering a changing steel fatigue resistance. In that case, the material parameters in the Lundberg-Palmgren equations are a function of the condition of the steel microstructure and of the internal operating temperature, but this is not explicitly stated.

In the work which follows, all of the aforementioned deficiencies or omissions in the Lundberg-Palmgren theory are addressed; a mathematical model for the prediction of fatigue failure, or the nonoccurrence thereof, in modern rolling bearing applications is presented. Moreover, a linkage between structural fatigue and rolling contact fatigue is established.

Mathematical Formulation of the Fatigue Failure Model

In rolling bearings, loads are transmitted from one ring to the other via rolling elements. The rotation of at least one ring and the resulting motion of the rolling elements induce a cyclic stress field inside the material of all the components. It is this cyclic stress field that is responsible for the fatigue of the rolling bearings. The fatigue is observed as the formation of a spall, and is considered to be terminated when the first such spall is created. Two phases are identified during the period leading to the formation of the spall: (i) the initiation phase, which begins with the cyclic stressing and extends up to the point when a macroscopic, self-propagating crack appears; (ii) the propagation phase, characterised by the crack propagation up to the complete formation of a spall. If the crack growth phenomenon is of interest, then this second phase can itself be sub-divided into two phases, an early crack growth and a final fracture phase. Here, however, as in previous work [2, 12], it is assumed that by far the major part of the bearing life is consumed during the initiation phase. Accordingly, the bearing life can be approximated by this initiation period and no consideration is given hereafter to the crack growth phase. This means that in endurance tests to verify the mathematical model, the number of revolutions to observance of the first spall is a sufficiently accurate measurement of fatigue life.

To calculate the fatigue life of a rolling bearing, the Weibull weakest link theory is used along similar lines to the Lundberg-Palmgren formulation [2]. Under such assumptions, the probability of survival ΔS_i of a volume element ΔV_i, sufficiently large to contain many defects, can be generally expressed as:

$$\ln\frac{1}{\Delta S_i} = F(N, \sigma_i - \sigma_{ui})\Delta V_i \tag{2}$$

where σ_i is a stress-related fatigue criterion (e.g., the orthogonal shear stress amplitude in reference [2]) and σ_{ui} a threshold value of the criterion; i.e., a fatigue limit, below which the volume element will not fail by fatigue. The fatigue criterion is introduced after considering the role of similitude between fatigue processes. It is therefore tacitly assumed that different fatigue processes are, or can be considered, identical when the criterion values are equal, regardless of any other differences in the stress tensor. In the present model, any choice of fatigue criterion is admissible as long as this criterion can be calculated from the components of the local stress tensor. Furthermore, it is now postulated that the function $F(N, \sigma - \sigma_u)$ can be directly approximated by power functions[1] similar to the ones used in reference [2] to calculate the probability of survival of the entire bearing; i.e.,

$$\ln\frac{1}{\Delta S_i} = A_i N^e H(\sigma_i - \sigma_{ui})(\sigma_i - \sigma_{ui})^c \Delta V_i \tag{3}$$

where

$$H(x) \text{ is the step function, i.e.,; } H(x) = \begin{matrix} 1 & x \geqq 0 \\ 0 & x < 0 \end{matrix}$$

An important difference between this formulation (equation (3)) and the Lundberg-Palmgren theory [2] is that the fatigue limit σ_{ui} is now introduced as a lower limit of the fatigue behavior; i.e., it is assumed that *no* failure can occur

[1] Power relations between fatigue life and stress, including a fatigue limit, have been used by Weibull [13] to approximate S-N curves. Moreover, similar power relations using a strain power term with a fatigue limit have also been previously used in reference [14], but again for an entire specimen.

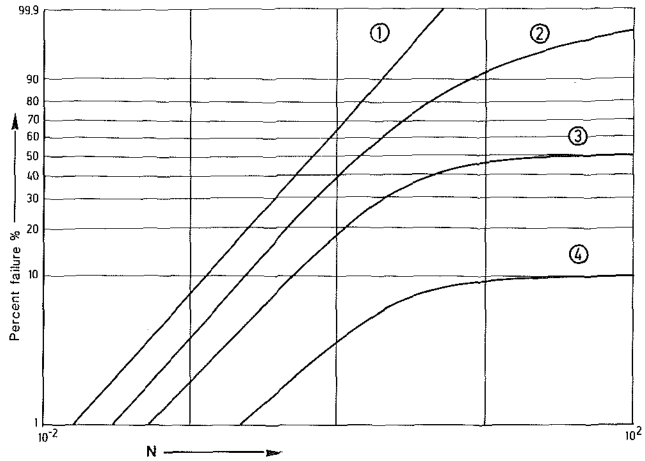


Fig. 5 Weibull plots for different I integral distributions

in this volume element if σ_i is less than or equal to σ_{ui} ($\sigma_i \leqq \sigma_{ui}$).

The two exponents e and c in equation (3) are assumed to be properties of the material and to remain constant throughout the material's load history. The two parameters A_i and σ_{ui} are assumed to be independent random variables. The variation of σ_{ui} is assumed to be bound by two limits, $\sigma_{u_{\min}}$ and $\sigma_{u_{\max}}$, where $\sigma_{u_{\min}}$ may be greater than zero, and its probability density function will be denoted hereafter as $g(\sigma_u)$. Before the probability of survival of an entire volume is computed, it is observed, in accordance with reference [2], that Weibull's theory is based on the assumption that the first crack leads to a break. In rolling bearings, however, examples of cracks which do not reach the surface seem to indicate that consideration ought to be given to the depths at which the most dangerous stresses occur. The practice which is adopted in the present work, as in reference [2], is to weigh the probabilities of each volume element, with a term $(z_i')^{-h}$, where z_i' is a stress weighted average depth. If the exponent $h = 0$, a life formula corresponding to Weibull's theory is obtained. The probability S_r that the entire volume V will endure N cycles is found by multiplication of the probabilities ΔS_i; i.e.,

$$S_r = \Delta S_1 \Delta S_2 \ldots\ldots \Delta S_n \tag{4}$$

and accordingly, for any given combination of the random parameters A_i and σ_{ui} in the n volume elements ΔV_i which make up the volume V, the probability of survival S_r can be found as

$$\ln\frac{1}{S_r} = N^e \sum_{i=1}^{n} A_i H(\sigma_i - \sigma_{ui}) \frac{(\sigma_i - \sigma_{ui})^c}{z'^h} \Delta V_i \tag{5}$$

Introducing $A_i = \bar{A} + \delta A_i$, equation (5) becomes

$$\ln\frac{1}{S_r} = \bar{A} N^e \sum_{i=1}^{n} H(\sigma_i - \sigma_{ui}) \frac{(\sigma_i - \sigma_{ui})^c}{z_i'^h} V_i$$

$$+ N^e \sum_{i=1}^{n} \delta A_i H(\sigma_i - \sigma_{ui}) \frac{(\sigma_i - \sigma_{ui})^c}{z_i'^h} \Delta V_i \tag{6}$$

In this equation, the summation term containing the variations δA_i is assumed small compared to the term containing the average $\bar{A}$ (remembering also that A_i and σ_{ui} have been assumed independent random variables), and therefore it is neglected as a first approximation. Passing to the limit, one obtains

$$\ln\frac{1}{S_r} = A N^e \int_{v_r} H(\sigma - \sigma_u) \frac{(\sigma - \sigma_u)^c}{z'^h} dV \tag{7}$$

where the integral is taken over the greatest volume V_r subject to risk of fatigue; i.e., the volume in which $\sigma > \sigma_{u_{\min}}$. For zero values of the integral in this equation; i.e., for stresses in the volume sufficiently low such that $\sigma \leqq \sigma_{u_{\min}}$ everywhere, the fatigue life of the volume V becomes infinite.

Denoting the random integral in equation (7) as I and its probability density as $f(I)$, the probability $P_0\{I \in [I_0, I_0 + dI]\} = f(I_0)dI$. The probability of survival S_0 when $I = I_0$ is from equation (7) $S_0 = e^{-AN^e I_0}$ and therefore the joint probability P_J of the two independent events, $I \in [I_0, I_0 + dI]$ and survival, is $P_J = P_0 * S_0 = e^{-\bar{A}N^e I_0} f(I_0)dI$. Furthermore, the probability of survival S for I varying between $I_{\min}$ and $I_{\max}$ is the sum of all probabilities P_J for $I_0 \in [I_{\min}, I_{\max}]$ because the occurrence of any two different values of I are mutually exclusive events. Therefore, finally,

$$S = \int_{I_{\min}}^{I_{\max}} e^{-\bar{A}N^e I} f(I)dI \tag{8}$$

Equation (8) does not represent, in general, a Weibull life distribution. Instead, this equation describes the aggregate fatigue lives of a population of bearings consisting of many groups, characterized by a unique value of I and occurring with frequency $f(I)dI$. Each of these groups has Weibull-distributed fatigue lives. Examples of the deviations of equation (8) from the Weibull distribution are shown in Fig. 5. All Weibull plots in Fig. 5 are computed for I distributions between 0 (which corresponds to infinite lives) and $I_{\max}$, where $I_{\max}$ is chosen for exemplification purposes as $I_{\max} = 1/\bar{A}$. Curve 1, the straight line, corresponds to a very narrow I distribution around $I_{\max}$; i.e., $f(I) \approx \delta_D(I_{\max})$, (where $\delta_D(x)$ is the Dirac function). This implies a unique value for I, which then results in a Weibull distribution (straight line). Curve 2 is obtained for a uniform distribution; i.e., $f(I) = 1/I_{\max}$. This curve deviates substantially from a straight line only at high failure percentages as the population now includes in the limit bearings with infinite lives. Finally, when a given fraction q of the overall bearing population belongs to a group with an infinite life ($I = 0$), while the rest, as before, have, for example, a uniform distribution, the combined distribution $f(I)$ can be expressed as $f(I) = q\delta_D(0) + (1-q)/I_{\max}$. Curves 3 and 4 in Fig. 5 represent the Weibull plots for $q = 0.5$ and $q = 0.9$, respectively. The early part of these curves approximates Weibull distributions (straight lines), while the later part tends asymptotically to the assumed infinite life fractions of survival q, as expected. The convex behavior of the Weibull plots at higher failure percentages, for which equation (8) allows, has a physical basis insofar as it has been observed in the past [15]. The issue of bearings attaining infinite fatigue lives will be discussed in detail in the section of this paper dealing with bearings.

The exponential part of the integrand in equation (8) can be approximated by $k+1$ terms of the McLaurin series, as follows:

$$e^{-\bar{A}N^e I} = \sum_{n=0}^{k} (-1)^n \frac{\bar{A}^n N^{ne} I^n}{n!} \tag{9}$$

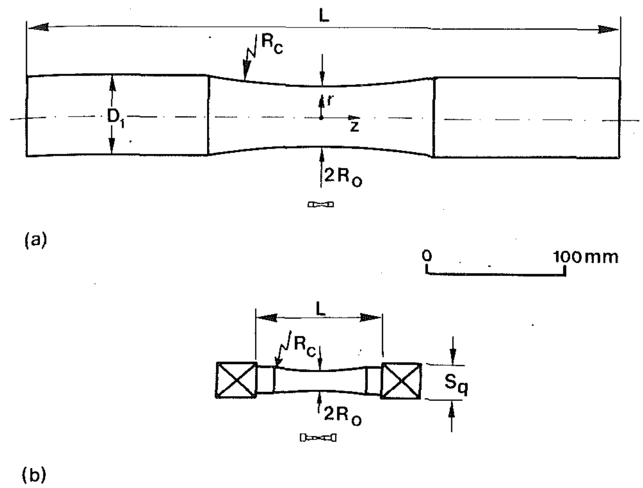
and consequently equation (8) becomes

$$S \approx \int_{I_{\min}}^{I_{\max}} \sum_{n=0}^{K} (-1)^n \frac{\bar{A}^n N^{ne} I^n}{n!} f(I)dI$$

$$= \sum_{n=0}^{K} (-1) \frac{\bar{A}^n N^{ne} \overline{I^n}}{n!} \tag{10}$$

where $\overline{I^n}$ are the moments of I.

The study of the bearing lives throughout the complete range of failure percentages, as described by equation (8) or



Dimensions of bending specimens, mm				Dimensions of torsion specimens, mm			
R_o	D_i	L	R_c	R_o	Sq	L	R_c
20.32	51.8	373.38	508.0				
17.78							
5.08	19.05	158.75	127.0	6.35	22.225	79.375	174.625
3.175	9.525	63.5	38.10				
2.794				2.794	9.525	34.925	76.2
1.27	4.715	34.29	25.4	1.27	4.775	15.875	34.925
0.635	2.388	17.02	12.7	0.635	4.775	15.875	17.272

Fig. 6 Dimensions of specimens used in reference [16]

equation (10), requires the knowledge of the distribution $f(I)$. For this, both the distribution of σ_u, $g(\sigma_u)$ and its autocorrelation $R_{\sigma u}(d)$, as function of distance d, are required. When these are known, or plausibly guessed, approximations of $f(I)$ can be obtained by sampling from the σ_u distribution. Such approximations of $g(\sigma_u)$ and $f(I)$ and their effect on fatigue lives is currently under investigation and will be subsequently reported. In bearing applications, however, low percentages of failure, typically of the order of 10 percent, are of interest. Then S may be approximated using the first two terms of the summation in equation (10) as

$$S \approx 1 - \bar{A} N^e \bar{I} \tag{11}$$

where $\bar{I}$ is the average of I.

The distribution implied by equation (11) is an approximation of the Weibull distribution

$$\ln \frac{1}{S} = \bar{A} N^e \bar{I} \tag{12}$$

a fact also confirmed by the straight, early parts of the plots of Fig. 5. From $g(\sigma_u)$, the distribution of σ_u, $\bar{I}$ can be obtained as

$$\bar{I} = \int_{V_R} \int_{\sigma_{u\min}}^{\sigma_{u\max}} H(\sigma - \sigma_u) \frac{(\sigma - \sigma_u)^c}{z'^h} g(\sigma_u) d\sigma_u dV \tag{13}$$

The exact Weibull distribution, equation (12), applies also to the *entire* range of failures when the material is considered homogeneous, so that $R_{\sigma u}(d) = 0$ when $d \neq 0$. In a sense, this implies that defects of infinitesimal size determine the σ_u value of an infinitesimal volume dV. This may be considered as a valid approximation for very "clean" steels containing only very small defects. Otherwise, as explained above, this formulation is valid only as a low failure percentage analysis.

Introducing a discretized form of integration for σ_u and substituting in equation (11), (or equation (12)), one obtains, after some manipulation,

$$\ln \frac{1}{S} \approx \bar{A} N^e \sum_{i=1}^{k} g_i \int_{V_{R_i}} \frac{(\sigma - \sigma_{ui})^c}{z'^h} dV = \bar{A} N^e \sum_{i=1}^{k} g_i \bar{I}_i \tag{14}$$

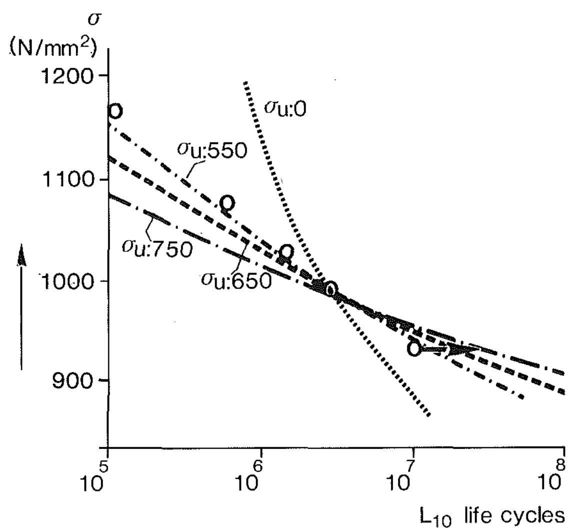


Fig. 7(a) Martensitic

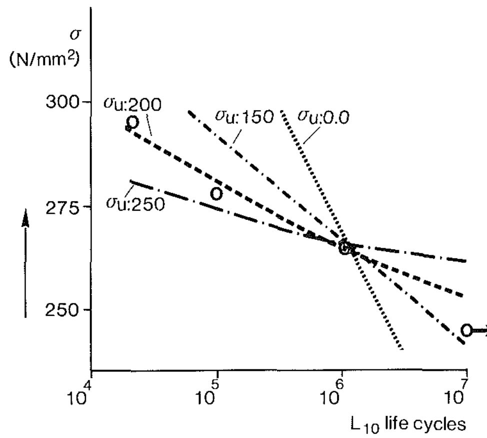


Fig. 7(b) Soft annealed

Fig. 7 Variation with stress of experimental and calculated L_{10} of rotating beams.
o experiment
calculation (with various σ_u's)
→ unbroken specimen (experiment)

where g_i represents each of the k fraction occurrences of σ_u, and the risk volume V_{R_i} corresponds to σ_{ui} (i.e., $\sigma > \sigma_{ui}$ in V_{R_i}). Each of the integrals $\bar{I}_i$, in the summation term, can be considered as the value of I for a volume which has a constant value of σ_u throughout. The summation term in this equation represents the average of these integrals. If the variation of σ_u can be neglected as an approximation for high quality homogeneous steel, equation (14) becomes

$$\ln \frac{1}{S} \approx \bar{A} N^e \int_{V_R} \frac{(\sigma - \sigma_u)^c}{z'^h} dV \tag{15}$$

This equation resembles the Lundberg-Palmgren formulation insofar as the integral over the risk volume now replaces the term $\tau_{0}^{c} z_0^{1-h} al$ of equation (37) in reference [2], when the orthogonal shear stress τ_{yz} is chosen as the fatigue criterion. This equation will be used as an approximation to compare predicted and experimentally obtained lives, not only for bearings, but also for other specimens (beams) used in fatigue tests. Unless stated, the same values of c, e, h, as in reference [2] will be used in these comparisons. This approximation is justified to some extent by the "cleaner" and more homogeneous steels used today in the manufacture of

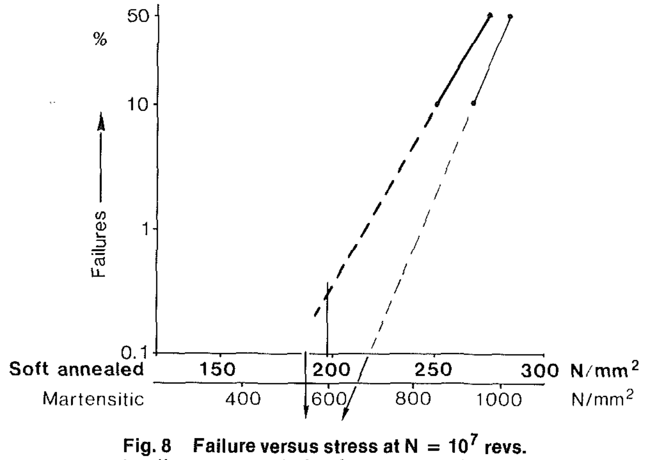


Fig. 8 Failure versus stress at N = 10^7 revs. (· · · linear extrapolation from test results).

rolling bearings. Such materials are expected to have smaller σ_u variability than steels used at earlier times (1930's) pertinent to the development of the Lundberg-Palmgren theory. The effects of the σ_u variability will also be indicated in a bearing example.

Applications

The fatigue model developed in the previous section is based on an empirical power law which has its origins in the Lundberg-Palmgren theory for complete bearings. For verification purposes, calculated fatigue lives using the present model are compared to experimental data obtained not only from bearings, but also from rotating beams, beams in torsion, and flat beam bending tests. Such comparisons using simple stress fields may also indicate wider applicability of the model to the study of fatigue initiation, since no limiting assumptions applicable only to rolling bearings were necessary in the formulation of the theory. The only requirement for the application of the present model is that the initiation stage of fatigue is much longer than the crack propagation stage. In addition, the material constants may have to be adjusted when different materials are used.

Rotating Beams. In the rotating beam tests, the stress field is uniaxial, and the alternating normal stress is the generally accepted fatigue criterion. The cross-section of the specimens varies, as shown in Fig. 6, and the magnitude of the alternating normal stress σ_z can be expressed in a polar coordinate system r, θ, z as

$$\sigma_z = \sigma_0 R_0^3 \frac{r}{[R_0 + R_c - \sqrt{R_c^2 - z^2}]^4} \tag{16}$$

where σ_0 is the maximum stress occurring at the surface of the smallest cross-section. Using equation (15), the L_{10} life of a rotating beam (for high cycle fatigue) can be expressed

$$\ln\frac{1}{S} = 2\pi \bar{A} N^e \int_{-z_l}^{z_l} \int_{R_l}^{R_s} \frac{(\sigma_z - \sigma_u)^c}{z'^h} r dr dz \tag{17}$$

where R_s is the variable surface radius and R_l and z_l, $-z_l$ are the radius and the z distances which delimit the volume at risk. The depth weighing factor z' in the radial direction is computed at each cross-section in the way explained in the previous section, that is, as a damaging stress-weighted average depth,

$$z' = \frac{\int_{R_l}^{R_s} (\sigma_z - \bar{\sigma}_u)(R_s - r) dr}{\int_{R_l}^{R_s} (\sigma_z - \bar{\sigma}_u) dr} = \frac{1}{3}(R_s - R_l) \tag{18}$$

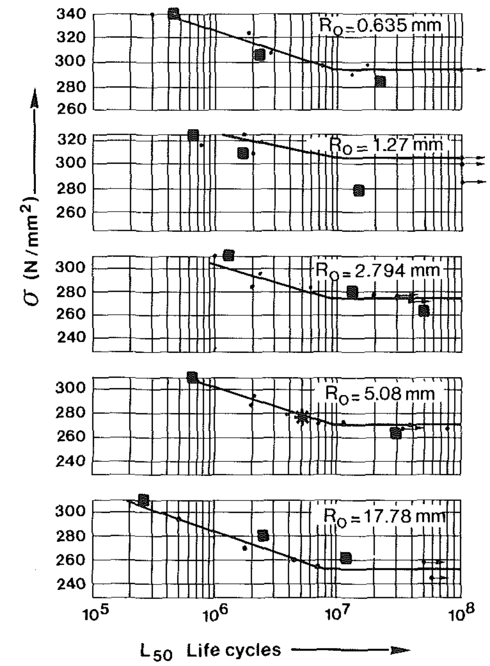


Fig. 9 L_{50} fatigue lives of rotating beams of various sizes; —◦ experiment (reference [16]), → unbroken specimen (experiment), ■ computed life (* reference point).

Comparisons between experimental data from beams of R_0 = 3.5 mm and R_c = 127 mm and computed results from equation (17) are presented in Fig. 7.

Two extreme AISI 52100 steel variants were used for these comparisons; a very hard martensitic and a soft annealed variant. In the comparisons, the exponents c and e were taken as 31/3 and 10/9 respectively, according to the Lundberg-Palmgren theory. These values were selected after some initial comparisons, where the sensitivity of the results to the variability of the constants was investigated. Experimental fatigue limit estimates were not available, and hence a parametric study using different values of the fatigue limit σ_u was conducted for each material variant. (Note that only two experimental points are required to obtain the two unknowns, $\bar{A}$ and σ_u.) The constant of proportionality $\bar{A}$ is obtained from one set of experimental values, σ and L_{10}, hereafter called the reference point, which is the common point of all calculated curves in Fig. 7. These values of $\bar{A}$ and σ_u are subsequently used to obtain the complete theoretical S-N curve. Figure 7(a) shows the L_{10} lives for the martensitic variant obtained from both the fatigue tests and the aforementioned calculatoin. Each of the calculated S-N curves corresponds to the indicated value of σ_u, and it can be seen from this figure that the best agreement is obtained for σ_u = 550 N/mm^2.

In a similar way, from Fig. 7(b) the best agreement for the soft annealed variant is obtained for σ_u = 200 N/mm^2. Furthermore, a plot at $N = 10^7$, with extrapolation at small probabilities of failure from the L_{10} and L_{50} experimental values, is shown in Fig. 8. Here it is shown that the values giving the best agreement between calculated and observed lives, i.e., σ_u = 550 N/mm^2 for the martensitic and σ_u = 200 N/mm^2 for the soft annealed variant, correspond to low probabilities of failure, namely 0.008 and 0.3 percent for each variant respectively. This suggests that the values of the best agreement in Figs. 7(a) and 7(b) may be suitable estimates of the fatigue limit for low failure percentages, or at least sufficiently close approximations for engineering purposes.

The agreement between experimental and calculated results using the chosen σ_u values for these beams is generally good in

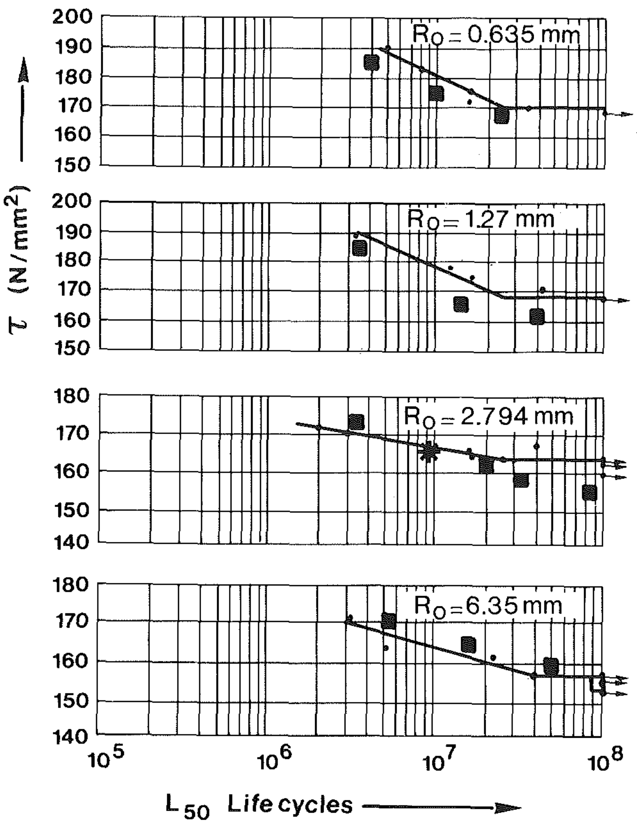


Fig. 10 L_{50} fatigue lives of beams of various sizes in torsion;
—∘ experiment (reference [16]),
→ unbroken specimen (experiment),
■ computed lives (* reference point).

the stress range examined. It should be noted, however, that at lower stresses (and consequently longer lives), this agreement may deteriorate as a result of the use of the simplified equation (15). In this case, as previously explained, higher moments of the random integral I may be required in the original equation (10) as higher percentages of the rotating beams attain infinite lives. Next, the effect of size was investigated. For this, the experimental results of beams, which were polished after stress relieving, were selected for comparisons from fatigue tests reported in reference [16]. Well-polished beams were selected, because in the modelling the stress field pertains to perfectly smooth beams. L_{50} lives were calculated for all the rotating beam sizes shown in Fig. 6. Computed and experimental results for these beams are shown in Fig. 9. The constant $\bar{A}$ and a fatigue limit of σ_u = 185 N/mm^2 were obtained from one beam size, R_0 = 5.08 mm, in the manner explained in the previous comparisons. Better agreement was obtained using c = 8 instead of c = 31/3, which may be more appropriate for plain carbon steel which was used to make these beams. This value of c was used for all comparisons in Fig. 9. The agreement between calculated and experimental lives for the entire size range appears reasonable.

Beams in Torsion. In addition to the rotating beams, tests of various sizes of beams in fully reversed torsion are also reported in reference [16]. The sizes tested are also indicated in Fig. 6. In this case too, the shear stress field can be obtained analytically and the S-N curves (L_{50} lives) can be calculated from the equivalent of equation (17), using the shear stresses as fatigue criterion. The same value of c, c = 8, was used as previously for the rotating beams made of the same steel. The fatigue limit was not obtained by a parametric study, but it was deduced from the tensile value σ_u as τ_u = 108 N/mm^2, using the von Mises equivalence. The constant $\bar{A}$ in torsion

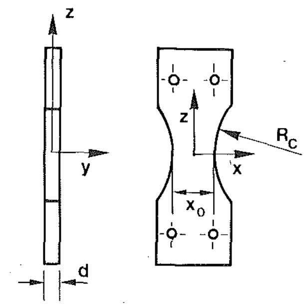


Fig. 11(*a*) Flat beam configuration

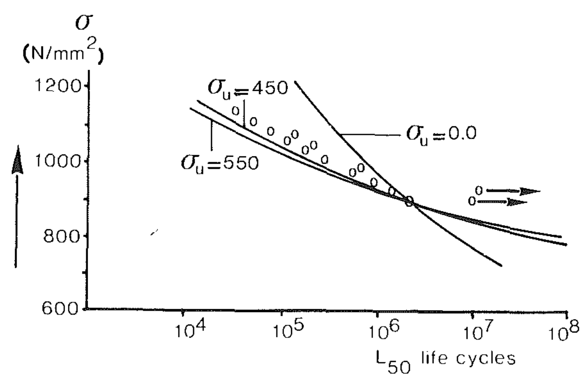


Fig. 11(*b*) L_{50} fatigue lives;
∘ experiment (reference 17),
→ unbroken specimen (experiment)
— computed lives for different σ_u values

was then obtained from one experimental point, the reference point shown in Fig. 10. Again, the agreement between experimental and computed lives is reasonable throughout the size range of the beams, especially for higher stresses.

Flat Beams in Reversed Bending. Finally, comparisons between a calculated and an experimental S-N curve were obtained for a flat beam made of bearing steel and tested in reversed bending [17]. In this case, the normal stress σ_z is used as the fatigue criterion. The configuration used in reference [17] is shown in Fig. 11, and the normal stress σ_z at any location is independent of x and can be calculated as

$$\sigma_z = \sigma_0 \frac{2y}{d} \frac{x_0}{x_0 + R_c - \sqrt{R_c^2 - z^2}} \tag{19}$$

In reference [17], an assessment of the effects of residual stresses induced by different rates of griding fatigue is made. The comparison of L_{50} lives shown in Fig. 11(*b*) pertains to beams with the smallest residual stresses produced by gentle grinding, since a full study of the other cases is beyond the scope of the present work. In this case, the small residual stresses relevant to this comparison were added to the stress field of equation (19). For the comparison shown in Fig. 11(*b*), the Lundberg-Palmgren values of the exponents c, e, and h were used, appropriate for bearing steels; the lives calculated with σ = 450 N/mm^2 show good agreement with the experimental values throughout the tested range of stress.

Fatigue Lives of Rolling Bearings. Finally, the fatigue live model described here was used to predict bearing lives. For this purpose, computed lives from the model were compared to the L_{10} lives obtained from endurance tests of a 6309 deep groove ball bearing (DGBB). The results of these tests (four in all), summarized in Table 1, are particularly suitable for comparisons. This is because identical, carefully selected, bearings made of two different steel qualities were tested under both excellent and marginal lubrication conditions (high and low Λ). The inner ring of the bearing is the failure component in all cases. Details of these and other com-

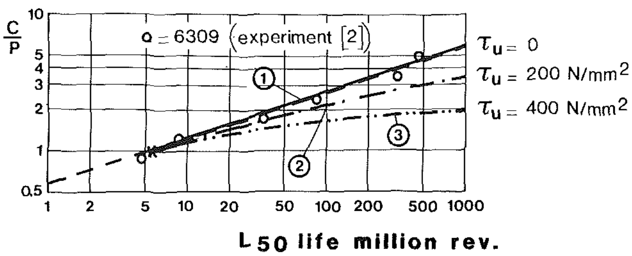


Fig. 12 Comparison of experimental and theoretical L_{50} ——, from reference [2] with the present model for different τ_u (* reference point)

Table 1 6309 ball bearing tests at radial load = 18835 N ($C/P = 2.8$)

	"Good" material (few oxide inclusions)	"Bad" material (many oxide inclusions)
Good lubrication (full lubricant film generation, $\Lambda^* > 10$)	$L_{10} > 1000$ Test discontinued	$L_{10} = 116$
Marginal lubrication (partial lubricant film generation, $\Lambda^* \approx 1.5$)	$L_{10} > 200$ Test discontinued	$L_{10} = 17.3$

$$\Lambda = \frac{\text{lubricant film thickness}}{\text{composite surface roughness}} = \frac{h}{\sigma}$$

parisons, all pertaining to the 6309 DGBB, will be given below, individually for each case. The subsurface stress fields used in all these comparisons were calculated from a special computer program based on the work of Kalker [18]. This program calculates the elastic field in a half-space due to an arbitrary load distributed over a bounded region of the surface. Sufficiently fine grids were used in planes normal to the rolling direction, under the contact area, so that no significant grid effects were present in the integrations. Whenever frictional stresses (traction) were introduced on the contact surface, these were calculated locally with their appropriate signs, taking into account the position of the two pure rolling axes induced by the curvature of the track. In the high Λ case, the coefficient of friction μ was taken to vary locally in proportion to the local pressure [19], with maximum values deduced from rotating disk experiments. In the case of low Λ, a constant $\mu = 0.15$ was assumed to characterize the lubricating conditions. Cubic mean ball loads, as in reference [2], were used for each bearing load.

First, 6309 DGBB L_{50} lives computed with the present model were compared to experimental and theoretical results for the same bearing from reference [2]. For this comparison, the assumptions of reference [2] were introduced in the present model, namely: (i) the subsurface stress is Hertzian (no surface friction); (ii) the fatigue criterion is the orthogonal shear stress τ_{yz}; (iii) the fatigue limit $\tau_u = 0$. Under these assumptions and using a reference point common to the theoretical curve of reference [2] at $C/P = 1$ to calculate $\bar{A}$, the theoretical results of reference [2] are reproduced almost identically in curve 1 of Fig. 12. Introduction of an increasing faigue limit τ_u in the model produces, as expected, increasingly higher lives, also shown in the same figure. Curves 2 and 3 in this figure, corresponding to $\tau_u = 200$ N/mm^2 and $\tau_u = 400$ N/mm^2, are plotted with the same reference point to demonstrate the progressive deviation from the straight line. In general, however, depending on the value of $\bar{A}$, these curves may be translated parallel to the life axis.

Next, comparisons with the experimental results of Table 1 were made. In this case, contact surface frictional stresses were introduced in the way described above. Then, the subsurface stress fields are different as compared to the Hertzian stress field. τ_{yz}/p_0 shear stress contours are shown in Figs. 13(a) and 13(b) at planes of $y/b = 0.8$ and $y/b = -0.8$

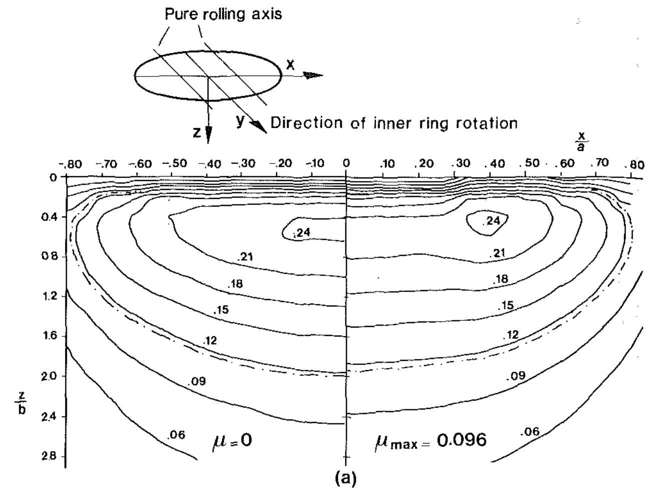


Fig. 13(a) Shear stress τ_{yz}/p_0 contours on a plane $y/b = -0.8$ with and without friction

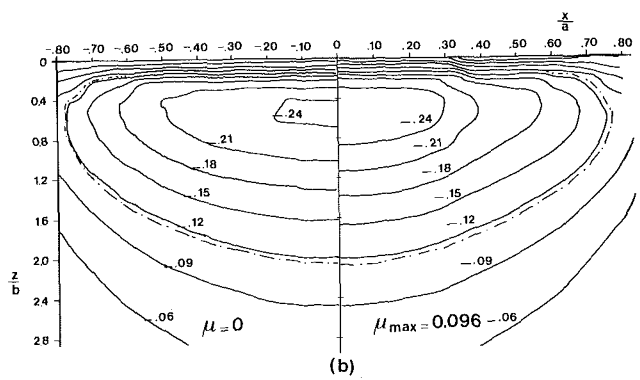


Fig. 13(b) Shear stress τ_{yz}/p_0 contours on a plane $y/b = 0.8$ with and without friction

for a bearing load $C/P = 2.8$. y is the direction of rotation of the inner ring, and the origin of the coordinate system is in the center of the contact ellipse, as shown in the key of Fig. 13. The left-hand side of both Figs. 13(a) and 13(b) correspond to the Hertzian field ($\mu = 0$), the assumption used in reference [2], while the right-hand side corresponds to variable μ case with $\mu_{max} = 0.096$ (high Λ). Similar shapes were obtained for $\mu = 0.15$ (low Λ). It can be seen from Fig. 13(a) that the reversal of frictional stresses across the pure rolling axes introduces an enclave of high shear stress away from the x-plane of symmetry ($x = 0$) at $y/b = 0.8$, a feature absent in Hertzian stress fields. At $y/b = -0.8$, Fig. 13(b), the maximum shear stress τ_{yz} occurs in the plane of symmetry ($x = 0$), but the pattern is somewhat different to the Hertzian stress field. The broken line in Figs. 13(a) and 13(b) demarkates the stress contour $\tau_{yz} = 300$ N/mm^2 and consequently the area in those cross-sections exposed to fatigue when $\tau_u = 300$ N/mm^2. Two different fatigue criteria are used to exemplify L_{10} computed lives with the present model: the shear stress amplitude τ_a and the von Mises equivalent stress σ_{VM}. In the case of τ_a, the passage of a rolling element (ball) induces a nonfully reversed shear stress cycle at each location as a result of the surface frictional stresses. The absolute maximum value of the shear stress τ_{max} in this cycle is assumed to modify the fatigue limit τ_u in each location. This modification is introduced in accordance with the findings of references [20] and [21] for plain carbon steels; τ_u is assumed to remain unchanged if τ_{max} does not exceed the elastic limit τ_e, and to diminish linearly to zero for τ_{max}, varying between τ_e and the fracture strength τ_f. Given the complexity of the subsurface stress field, it is clear that the most damaging cycle at a point will not necessarily occur in

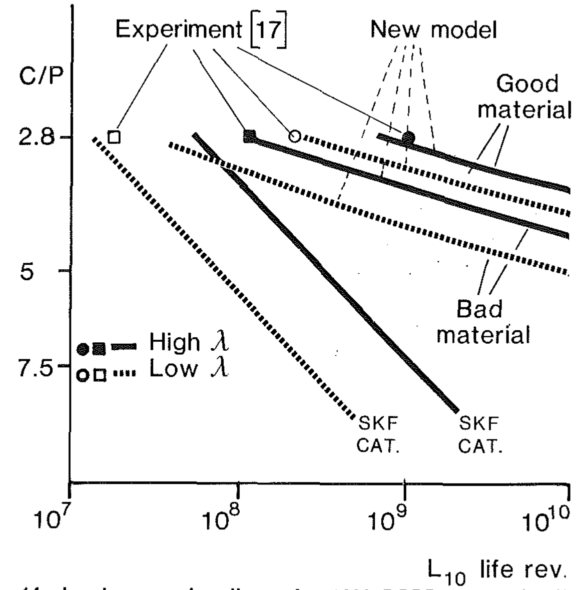


Fig. 14 Load versus L_{10} lives of a 6309 DGBB made of different materials and under different lubrication. τ_a (shear stress amplitude) used as fatigue criterion. ○, ●, □, ■ experimental values. "Good" material experimental values ○, ● represent lowest possible.

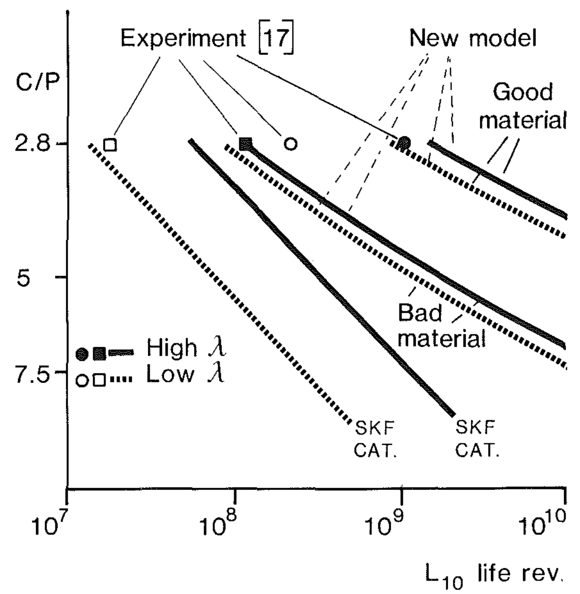


Fig. 15 Load versus L_{10} lives of a 6309 DGBB made of different materials and under different lubrication. σ_{vM} used as fatigue criterion. ○, ●, □, ■ experimental values. "Good" material experimental values ○, ○ represent lowest possible.

planes parallel to the contact surface (where τ_{yz} stresses lie). The variation of the subsurface stress tensor is investigated at each grid point, and the most damaging shear stress cycle ($\max(\tau_a - \tau_u)$) is obtained. These shear stresses almost exclusively lie in planes parallel to the axis of the bearing, but their inclinations to the surface vary with depth z and position x. Using these values, the probability of survival of the inner ring for a non-distributed τ_u is

$$\ln\frac{1}{S} = \bar{A}N^e \int_{B_r}\int_l \frac{(\max(\tau_a - \tau_u))^c}{z'^h}\, dl\, dB \qquad (20)$$

where dB is the incremental area of the cross-section of the ring, B_r signifies the part of this cross-section where $\max(\tau_a - \tau_u) > 0$, and dl is the increment in the rolling direction. From admittedly incomplete data and a lot of guesswork, τ_u for the "bad" and "good" materials are introduced as non-random variables of τ_u = 300 N/mm^2 and τ_u = 350 N/mm^2

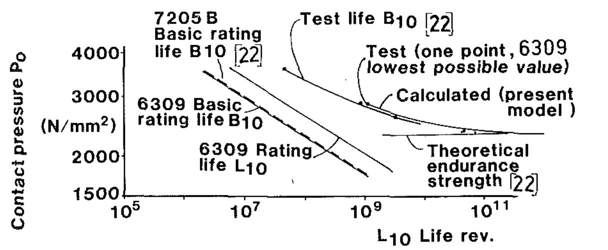


Fig. 16 Variation with contact stress of rating, experimental and theoretical L_{10} lives of a 7205 B angular contact ball bearing and a 6309 deep groove ball bearing. ● experimental values

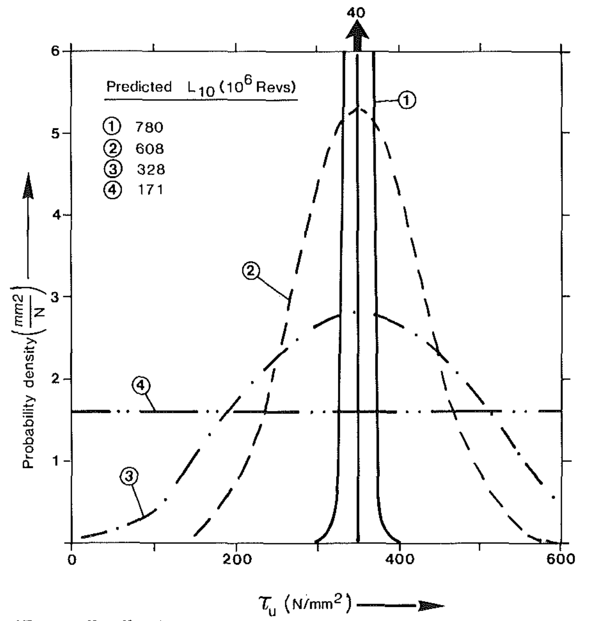


Fig. 17 τ_u distributions and corresponding L_{10} lives of a 6309 DGBB at C/P = 2.8 (good material – high Λ)

respectively, and a ratio of $\bar{A}_B/\bar{A}_G \approx 5$ is assumed. For this reason, what follows should be considered more of an illustration of the way that the new model can be applied to practical problems rather than a strict verifying comparison. Using one experimental datum, the L_{10} life for the "bad" material at high Λ, $\bar{A}_B$ can be obtained from equation (20). Then four theoretical "S-N" curves can be obtained describing the four tested combinations (Table 1) under varying load, Fig. 14. (Note that $\bar{A}_G$ cannot be obtained from a similar practice because no exact L_{10} lives are available for this material.)

In addition to these curves, the L_{10} (bearing catalogue) curves for the two lubrication conditions are shown in Fig. 14. The agreement between experiment and predictions, using the new model at C/P = 2.8, seems rather good. In contrast, the catalogue lives are in disagreement with experimental values, being substantially lower, especially for the good material. Furthermore, the predicted lives using the new model increase much more rapidly with diminishing loads as compared to the catalogue rating lives. Similar results are obtained using the von Mises equivalent stress as the fatigue criterion, and these are shown in Fig. 15. In this case, fatigue limits of σ_{vMu} = 519 N/mm^2 and σ_{vMu} = 606 N/mm^2 are used for the "bad" and "good" materials. Similar trends as in the previous case (Fig. 14) are observed. Presently, however, the increase in friction (low Λ as compared to high Λ condition) appear to influence (diminish) the difference in predicted lives less than in the previous case. Also, the predicted L_{10} life increase for diminishing loads appears to be less rapid when compared to the previous case, but is still more rapid than the catalogue predictions.

For comparison purposes, the 6309 DGBB calculated lives, for good material and high Λ, using τ_a as the fatigue criterion, are plotted in Fig. 16 together with the published experimental lives for the angular-contact ball bearing, 7205 B [22]. Moreover, in this figure, two rating L_{10} life curves are shown, the common basic rating straight line and the 6309 adjusted rating life. The lives are plotted versus contact pressure p_0, and similar shifts of the experimental B_{10} lives of the 7205 B bearing and the L_{10} lives predicted by the new model are observed, as compared to the rating lives. In addition, the slope of the experimental curve for the 7205 B bearing is in good agreement with the slope of the calculated curve using the present model. (The results, obtained using the von Mises equivalent stress as the fatigue criterion, show a very similar trend, but they are not shown.) This indicates a possible increase of the load/life exponent with diminishing loads.

Finally, the effects of the fatigue limit distribution on predicted lives are shown in Fig. 17. In this case, life predictions using equation (14) are obtained for the four τ_u distributions shown in Fig. 17. These are assumed to be truncated Gaussian distributions, with τ_u varying between $\tau_u = 0$ and $\tau_u = \tau_y$ (τ_y is the yield point). All these distributions have the same Gaussian mean, $E_G\{\tau_u\} = 350$ N/mm^2 the value used for good material, but different standard deviations. (Note that both the average and the standard deviation of the truncated distributions may be different as compared to those of the original Gaussian distribution as a result of the truncation.) Distribution 1 in Fig. 17 corresponds to $g(\tau_u) \approx \delta_D$ (350), which is equivalent to the assumption of a unique τ_u ($=350$ N/mm^2) used to obtain the good material lives in Fig. 14. Using the high Λ case for illustration purposes, where $L_{10} = 780\ 10^6$ revs., the L_{10} lives obtained for each of the other distributions are shown in Fig. 17. For these calculations, the τ_u range $(0-\tau_y)$ was discretized into 30 intervals. It can be seen there that the life reduction as a result of the spread of τ_u's can be substantial. This may be an important factor in predicting lives, especially for bearings fabricated from "less clean" materials where larger τ_u or σ_u spread is expected.

General Remarks. The use of a nondistributed fatigue limit (which was introduced in the comparisons in this section) implies that the "S-N" curves (or C/P versus N) for the various probabilities of survival and for diminishing loads, approach asymptotically the same horizontal straight line; i.e., the line of "infinite" lives. This line for a non-zero σ_u corresponds to non-zero stress or load. The same behavior is exhibited by the Lundberg-Palmgren model [2], but therein the "infinite" lives line corresponds to zero load. It may be argued that, should σ_u be a value close to zero, the exact position of the "infinite" lives is of academic interest only, since endurance tests to prove it either way will also have to be of "infinite" duration. With a gradual increase of σ_u, however, the traditional "knee" of an S-N curve may enter the region of practical interest. It is also interesting to note that improvements in the manufacture of rolling elements could themselves *appear* as an improvement of σ_u. This is because in reference [2], as well as in the present model, an "ideal" bearing stress field is used to compute the probability of survival. Any increase (local or global) of the stress field $\delta\sigma$, as a result of manufacturing errors, can be introduced in the $(\sigma-\sigma_u)$ term as $(\sigma+\delta\sigma-\sigma_u)$. However, this can also be written as $(\sigma-(\sigma_u-\delta\sigma))$, which exemplifies the point that σ_u now appears smaller.

In general, with an increase of σ_u, the load/life exponent in the rolling bearing may become different to 3 and variable with load. Such behavior is accounted for by the present model, and it may be argued that it is exhibited in the experiment of Zwirlein and Schlicht [22]. If the precise behavior near the fatigue limit is of interest, a more general form of the model, equation (8), will have to be used, as explained before. Finally, variation of material parameters or stresses with the number of cycles N is also admissible within the scope of the new model. Then, an additional integration over the N domain along the lines indicated in equation (25) or equation (32) of reference [2] is necessary. The development of residual stresses and of microstructural alterations observed under the raceways [11, 22], may be treated in this manner. Such effects, however, were not explicitly accounted for in the foregoing comparisons between experimental and predicted bearing fatigue lives. This is because of the lack of precise such experimental data in these endurance tests. Moreover, the material parameters can be introduced as functions of the internal operating temperatures, to account for the temperature effects on fatigue lives. The desirability of such refinements, which require very complex imput to the model, can only be established, however, after extensive comparisons of experimental and predicted lives.

Conclusions

In this paper, a new mathematical model for fatigue life prediction in rolling contact bearings and other fatigue-prone machine elements is introduced. It is shown capable of accounting for differences in the material through a fatigue limit (and its distribution) and for differences in the subsurface stress fields through the detailed usage of these fields. It is therefore possible with this new model to account for many phenomena, hitherto absent in the formulation of the fatigue life prediction of rolling bearings, e.g., contact surface friction. Moreover, it was also shown that this new model yields results identical to the Lundberg-Palmgren theory for the 6309 DGBB under the same assumptions, a fact indicating its flexibility.

Finally, it should be noted that the wider applicability of the exponents c, e, and h, "borrowed" from reference [2], the choice of the most suitable fatigue criterion, the related fatigue limit values and their distributions, are matters that can only be settled after examination of a wide variety of fatigue endurance tests of rolling bearings and other applications.

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